

ELEMENT C

Priority	Requirement	Prior Research Connection	Stakeholder Validation
#1 (High)	The materials of the device must be waterproof and be able to hold at least 500 ml of water to ensure the water doesn't leak out from the solution.	Based on research in element A, moisture can cause mold growth and pose potential health risks, according to the Property Casualty Alliance of Texas. Therefore, the material must not be able to absorb or obtain any form of water to prevent any increase in moisture in the environment. On the other hand, leakage of water can also create slippery walkways, which makes it very dangerous for pedestrians.	This solution was presented to one of the students who spent more than one class period at G-building daily. He estimated that a capacity of 500ml is probably not enough for capturing all the condensate produced in one day. Therefore, the design must either be cleaned multiple times a day, or increase in size in order to maximize the capacity to at least 750ml.
#2 (High)	The device must occupy an area smaller than 2 feet x 1 foot per AC unit so the water doesn't contact too many heated surfaces and evaporate rapidly into water vapor to increase the air humidity of the surrounding area.	According to previous research in element B, some of the potential solutions fit into this requirement. For instance, the drain trap only occupies 0.848 cubic feet of the space, and the area of the drain pan can differ depending on the size of the material. However, some require complex construction and require large space, such as the drop spillway, which would probably occupy more than 500 square feet.	This requirement was validated by one of the teachers who is assigned to the classroom. He states that the overall temperature of the building is about 80°F throughout most of the seasons, so the materials are easily heated under such a temperature. Therefore, by decreasing the contact area between condensate and the heated surface of the device, less evaporation could occur within or nearby it.
#3 (High)	The solution must be able to be moved, replaced, or uninstalled in less than a minute so the device is convenient for the user and they can clean the device easily in their daily routine.	In the prior research of element B, most of the attempted solutions satisfy this requirement except for the drain spillway, since it is solidly installed on the ground. The drain trap pipe has a cap, which allows the user to clean it with a brush. Similarly, the overflow safety switch also consists of a set of pipes; however, its maintenance might require the uninstallation of the pipe, but it might also be easy to achieve as well.	This requirement was validated by the janitor who is responsible for the cleanliness of the G-building surroundings; he confirmed that any solution that fits into this requirement would increase his efficiency in cleaning the walkway by concentrating all the condensate into one device and shortening the time it needs to remove moisture.
#4 (High)	The solution must be capable of collecting at least 90% of the condensation water produced by the AC unit and redirecting it into a contained area or container so that no water drips onto the walkway surface during normal operation.	According to prior research in element B, few potential solutions can meet this requirement. For example, rubber tubing can collect dripping water and channel it through tubing into grass or another container nearby. Another possible solution is a slanted collection tray, which allows water to naturally flow into a drainage tube using gravity. These systems work effectively because condensation typically forms in small but continuous amounts, making controlled collection possible without large equipment.	This requirement was validated by one of the teachers who is assigned a G-building classroom. He agrees that water should not contact the ground at all, and the capacity should be maximized as much as constraints and conditions allowed.
#5 (High)	The design can not have a blockage, causing the water line to go higher than 5" when passing through the design.	In element A, part 1's Research, when a condensation line is clogged with algae or debris, the AC is unable to hold the water and leads to the water leaks we see at the classroom's ACs. And in Element A's part 2, the water leaking can cause health risks to the primary users who walk past or are in	One student who accesses the G-building walkway comments that 5" of water is probably not a proper standard since it's still too much for the device to hold. Even if it is in a 1" wide pipe, 5" height of water would

		the area of the collected water, causing nausea, dizziness, headaches, and sometimes unconsciousness when the water is unable to be moved somewhere.	still make the walkway slippery and dangerous. A more proper standard might be 2.5".
#1 (Low)	The design must have an emergency system for when it gets clogged after not getting cleaned for a maximum of 3 days due to debris buildup.	In element B, prior design attempt #2, the AC condensate overflow safety switch allows the drip pan to be clogged and be filled with water, and a system triggers a signal for the AC to drain at a certain level. When applied to our design, this allows the custodians not to regularly clean the AC when it is a windy day, and debris can fall into our design.	This requirement was validated by the janitor who is in charge of maintaining the environmental cleanliness of G-building. He reminds us that sometimes, especially during long holidays, the walkways are usually not cleaned, so it is very possible that the situation in the requirement could happen. Having that kind of feature for our solution might do a little bit of help for him if the solution itself cannot block debris.
#2 (Low)	The design must lead to the grassy area outside of the classroom in the event of an overflow of water; the tube needs to carry water that drips from the AC to that grassy area in the G-building and K-building area.	In Element B's research, Prior design attempt #3, the rubber tubing allows the AC to transfer its refrigerant to somewhere else. We can add a draining system along with the rubber tubing to the designated area that will transfer the water. The draining design needs to be added due to the odd area where the water will drip from the AC unit. The rubber tubing allows for flexibility like a cable, for stakeholders to walk on without being bothered and bend the tube.	This solution was presented to a student who takes a class in that classroom four times a week. She suggests that if this requirement is fulfilled, the device can occupy the walkway in order to lead water from the AC units all the way to the grass, which violates one of the constraints for this project because it affects people's access to the walkway.